

SUPPLEMENTARY DATA

Maternal-fetal detection of rabies virus in the vampire bat *Desmodus rotundus*, Brazil

Data derived from the results presented in this study are available through the repository <https://github.com/mariguilardi/rabies>

This link will provide you with full access to:

- **Global alignments and phylogenies**
- **GenBank accession codes used in the datasets**
- **Genome Detective summary of each virus per pool**
- **SNP alignment and table**
- **MCC tree**
- **Supplementary Figures and Tables**

Material and Methods

Bat sampling and RABV screening

We sampled eleven pregnant females from four bat species: *Artibeus lituratus*, *Desmodus rotundus*, *Eumops perotis* and *Molossus molossus* (Table S1). For each individual, we collected the central nervous system (CNS), gastrointestinal tract (GIT), kidney and the corresponding fetus. Specimens originated from localities within Ceará State, Brazil, and were obtained between March 2023 and January 2024. Samples were acquired through passive surveillance conducted by the Brazilian National Rabies Prevention Program in collaboration with the Central Public Health Laboratory of Ceará (LACEN-CE). LACEN-CE performed the initial screening for rabies virus (RABV) using WHO-recommended fluorescent antigen test (FAT) and mouse inoculation test (MIT) techniques on CNS tissues. Ethical approval was granted by the Biodiversity Information and Authorization System (SISBIO no. 85149-1) and the UNIFESP Animal Experimentation Ethics Committee (protocols 329908092 and 8054091123).

Viral screening

Samples were extracted using the Quick-RNA™ Viral Kit (Zymo, California, USA) and subsequently subjected to cDNA synthesis with the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, California, USA), following the manufacturers' instructions. PCR assays were performed for each sample targeting viruses of public health relevance, including pan-Coronavirus (RdRp gene) [1], pan-Flavivirus (UC1 gene) [2], pan-Influenza (PB1 gene) [3], pan-Lyssavirus (N gene) [4], pan-Paramyxovirus (NS5 gene) [5] and pan-Rhabdovirus (L gene) [6]. PCRs were conducted using forward and reverse primers at 10 mM for each target and PCR

SuperMix (Invitrogen, USA). Amplicons were purified with the DNA Clean & Concentrator®-25 Kit (Zymo Research, USA) and Sanger sequenced at Genoma USP (São Paulo, Brazil) using an ABI 3730 DNA Analyzer platform. Preliminary taxonomic identification was performed using NCBI BLAST, considering the highest percent-identity score.

RNA-Seq

The samples were organized into pools based on the viral screening results (Table S2). Each sample was centrifuged to obtain the supernatant, which was subsequently filtered through 0.45 µm and 0.22 µm filters. Total RNA was then extracted and purified using the Quick-RNA™ viral kit (Zymo Research, USA). Total RNA samples were submitted to Genoma USP (São Paulo, Brazil) for Total RNA-Seq processing. Libraries were prepared using the RiboFree Total RNA Library Prep Kit (Zymo Research, USA), following the manufacturer's instructions. Briefly, total RNA was reverse-transcribed into cDNA while ribosomal RNA was simultaneously depleted. Partial P7 adapter sequences were ligated to the 3' ends of the cDNAs, followed by second-strand synthesis and ligation of partial P5 adapters to the 5' ends of the resulting double-stranded DNA molecules. Libraries were then amplified by PCR to incorporate full-length adapters. Library integrity and fragment-size distribution were assessed using Agilent's D1000 ScreenTape Assay on a TapeStation system. Finally, RNA-seq libraries were sequenced at Genoma USP on the Illumina NovaSeq 6000 platform (Illumina, USA) with a minimum depth of 60 million paired-end reads (150 bp) per sample.

Genome/Contig Assembly

The reads were processed with Genome Detective [7] for contig assembly and for the identification of viruses not targeted in the initial screening. For this analysis, we used the Pan-viral (2.14.6) pipeline for paired-end reads. This workflow filters high-quality reads, performs trimming when necessary, selects candidate viral reads, assembles contigs and ultimately generates consensus sequences by aligning contigs to reference genomes. The pipeline incorporates several tools, including FastQC (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>), Trimmomatic (<https://github.com/usadellab/Trimmomatic>), DIAMOND (<https://github.com/bbuchfink/diamond>), metaSPAdes (<https://github.com/ablab/spades>) and the Annotated Genome Aligner (AGA; <https://www.genomedetective.com/app/aga>). Assemblies identified as viral contigs by Genome Detective were further evaluated using NCBI BLAST to confirm their taxonomic identity, based on the highest percent-identity hit. ORF prediction and genome annotation were performed with NCBI ORFfinder (<https://www.ncbi.nlm.nih.gov/orffinder/>) and BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). Functional domain characterization for each ORF was carried out using searches in the Conserved Domain Database (CDD) (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>).

Recombination analysis

To assess the presence of recombination in the genomes, the RABV sequences were analyzed using RDP v.5 [8]. This *in silico* approach tests for recombination breakpoints by comparing each viral genome with its corresponding reference sequences retrieved from NCBI GenBank. The analysis incorporates six detection methods: RDP, GENECONV, Chimaera, MaxChi, BootScan and SiScan, and results were considered significant at $p \leq 0.05$.

Illumina Viral Surveillance Panel v.2 (VSP2)

We submitted the GIT and fetal tissues from the sampled individuals to a viral panel analysis using the Illumina Viral Surveillance Panel v2 Kit (VSP2) and the NextSeq 1000/2000 P2 XLEAP-SBS Reagent Kit. This panel employs target enrichment through a hybrid-capture method, enabling the detection of more than 200 viruses.

Total viral DNA and RNA were extracted using the Extracta® Kit – DNA e RNA de Patógenos MDx (Loccus, Brazil), following the manufacturer's instructions. The samples were then sent to Instituto Adolfo Lutz (Brazil) for library preparation and sequencing. For library preparation, RNA was first denatured and converted into cDNA. During the tagmentation step, the cDNA was fragmented and adapter sequences were added. Purification of the tagmented libraries was performed using magnetic beads. Libraries were then quantified and normalized, after which they were pooled for enrichment, with each pool containing one to three libraries. Capture probes were added to the pooled libraries to target regions of interest. Hybridized fragments were captured with magnetic beads, nonspecific binding was removed, and the enriched libraries were eluted. The enriched material was amplified through a 14-cycle PCR and subsequently purified using magnetic beads. Library quality was assessed on a Qubit platform (Thermo Fisher Scientific, USA). Sequencing was performed on a NextSeq 2000 platform (Illumina, USA).

Molecular confirmation of viruses identified through de novo assembly

PCR assays were performed to confirm the presence of Adenovirus (AAV), Hepatovirus (Roseolovirus genus), Human papillomavirus (HPV) and Mammalian orthoreovirus (MRV), which were detected by Genome Detective. Primers were

designed (Table S4) using Primer3Plus (<https://www.primer3plus.com/index.html>), with primer quality assessed using PCR Primer Stats (https://www.bioinformatics.org/sms2/pcr_primer_stats.html). *In silico* PCR validation was conducted using the *in silico* PCR amplification tool (<http://insilico.ehu.es/PCR/>). For the detection of Human betaherpesvirus 6, we used primer sets 7(2), 6(3) and 7(3) from the literature [9]. PCR assays were performed using forward and reverse primers at 10 mM for each target and PCR SuperMix (Invitrogen, USA). Amplicons were purified with the DNA Clean & Concentrator®-25 Kit and Sanger sequenced. Taxonomic identification of the resulting sequences was confirmed using BLAST.

Bayesian phylogeny and minimum spanning tree analyses

For the RABV genomes and contigs obtained from the *Desmodus rotundus* individual ID 321 (CNS, GIT, kidney and fetus), we aligned them with full-length genome sequences from GenBank (maximum length 12,000 bp) that contained metadata on host species, sampling date and geographic origin. Duplicate entries were removed using Sequence Cleaner (https://biopython.org/wiki/Sequence_Cleaner). Sequence alignment was performed with MAFFT [10]. The curated alignment was analyzed in BEASTX [11] to infer the evolutionary history and spatial diffusion patterns of the viruses using a Bayesian framework.

We implemented a two-step strategy to ensure proper mixing and convergence of the Markov chain. In the first step, we ran a short exploratory analysis (1 million generations) to estimate tuning parameters required for efficient proposal mechanisms during MCMC sampling. This preliminary analysis used the GTR substitution model with gamma-distributed rate heterogeneity (four categories), an uncorrelated lognormal relaxed molecular clock (UCLN), a constant population-size coalescent prior and Bayesian Stochastic Search Variable Selection (BSSVS; asymmetric) to infer

directional transitions among discrete geographic locations. In the second step, the optimized tuning parameters from the preliminary run were applied to a longer analysis of 150 million generations under the same model configuration. This extended run allowed us to obtain stable posterior estimates of evolutionary rates, divergence times and phylogeographic transitions. Convergence and effective sample sizes (ESS) were assessed using Tracer, ensuring reliable parameter estimation across the posterior distribution.

Maximum Likelihood and Maximum Parsimony analyses

Maximum-likelihood phylogenies were reconstructed for each virus identified. For RABV, we generated two separate phylogenies. The first used sequences from the L and N genes, which were Sanger sequenced, together with homologous sequences retrieved from GenBank. Sequences for each gene were individually aligned and subsequently concatenated. The second phylogeny was based on the RNA-seq data, in which we aligned all RABV sequences obtained in this study (full genomes and contigs) together with the full genomes used in the Bayesian analysis. Branch support for all RABV phylogenies was assessed using the approximate Bayes test (13). For AAV, Hepatovirus, HPV and MRV, we retrieved full genomes available in GenBank and aligned them with the contigs generated in this study. All alignments were performed using MAFFT (10). Maximum-likelihood phylogenies were then inferred with IQ-TREE 2.4 [12] under a GTR+G model, employing a combination of nearest-neighbor interchange (NNI) and subtree-prune-regraft (SPR) search strategies. Branch support was again evaluated using the approximate Bayes test [13].

To infer the ancestral states associated with our samples, we performed a Maximum Parsimony analysis of Simulated Characters in Mesquite 4.01 [14], using the topology recovered from the Bayesian phylogeny. This approach allowed us to

evaluate the most parsimonious reconstruction of trait evolution by generating simulated character histories across the fixed tree and identifying the character-state distributions that minimized the number of inferred evolutionary changes. This analysis provided an additional framework for interpreting lineage relationships and potential host-tissue associations within our RABV dataset.

Single-nucleotide polymorphism analysis

We estimated single-nucleotide polymorphisms (SNPs) in the RABV sequences recovered from the tissues of *D. rotundus* (ID 321) using Geneious R7 [15]. To evaluate SNP robustness under different conditions, we conducted the analyses using an alignment generated with MAFFT (10). We aligned the genomes and contigs obtained from the CNS, GIT, kidney and fetus of *D. rotundus* (RNA-Seq data) together with two reference sequences from the Bayesian phylogeny (GenBank IDs KX148109 and EU293113). We applied four reference-based SNP calling approaches: 1) KX148109 as the reference sequence; 2) EU293113 as the reference sequence; 3) the RNA-seq CNS genome as the reference sequence; and the 4) RNA-seq fetal genome as the reference sequence.

SNPs were called using a minimum variant frequency threshold of 0.001 and a maximum variant p-value of 10^{-6} , identifying variants both within and outside coding regions. SNP distributions across the RABV genome were visualized in R 4.3.3 (<https://cran.r-project.org/>) using ggplot2 (<https://ggplot2.tidyverse.org/>) and ggbreak (<https://cran.r-project.org/web/packages/ggbreak/vignettes/ggbreak.html>).

Coverage depth and viral abundance between tissues

We retrieved the RABV BAM files generated by Genome Detective for the RNA-seq reads obtained from the CNS and fetus of *D. rotundus* (ID 321). Reads generated

using the VSP2 panel for the fetal tissue were also submitted to Genome Detective to obtain the corresponding RABV BAM file. All BAM files were processed with samtools [16] to extract coverage depth at each nucleotide position along the RABV genome. We also extracted from Genome Detective the total number of reads, the number of viral reads and the number of RABV-specific reads identified for each RNA-sequenced tissue (CNS, GIT, kidney and fetus). These values were used to estimate the relative abundance of viral reads across tissues. Coverage profiles were visualized in R v.4.3.3 using ggplot2 and ggbreak packages, allowing the detection of uneven sequencing depth, potential coverage dropouts and highly supported genomic regions.

Data availability

The Sanger sequences for the CNS of *Desmodus rotundus* ID 321 were previously published [17] under the Genbank accession codes PP389347 (polymerase gene) and PP389310 (nucleoprotein gene). The Sanger and NGS sequences produced in this work were deposited in GenBank under BioProject PRJNA1165772 with the BioSample accession codes described in Table S5.

Supplementary Figure Legends

Figure S1. RABV dynamics between mother and fetus. A) Parsimony-based ancestral state reconstruction depicting the evolutionary relationships of RABV sequences recovered from multiple tissues. Green branches (upper right) highlight RABV detected across distinct biological compartments clustering with a canine-derived lineage (GenBank ID KX148109), illustrating continuous viral gene flow and intra-host dissemination. B) Single-nucleotide polymorphisms (SNPs) identified in alignments of RABV isolates from CNS, GIT, kidney and fetus, compared with GenBank references EU293113 and KX148109 (reference). Shapes indicate the tissue of origin for each SNP.

Figure S2. Maximum-likelihood phylogenies reconstructed for RABV and other viruses identified by RNA sequencing. Samples from *Desmodus rotundus* ID 321 are shown in red, and samples from non-hematophagous bat species are shown in blue. One asterisk indicates node support ≥ 0.75 ; two asterisks indicate node support ≥ 0.95 . A) RABV clades reconstructed from complete genomes obtained from CNS, GIT, kidney and fetal tissues of *D. rotundus* (Pools 1-4) and from CNS samples of *Molossus molossus* (Pools 5, 8 and 11). B) Phylogeny based on concatenated L and N genes, including isolates from *D. rotundus* (CNS, kidney and fetus) and *M. molossus* (CNS and anal swab). C) Adeno-associated virus (AAV) phylogeny based on sequences recovered from *D. rotundus* (CNS, GIT, kidney and fetus; Pools 1-4). D) Hepatovirus (Roseolovirus genus) phylogeny constructed from contigs recovered from the GIT and fetal tissues of *M. molossus* (Pools 12 and 14). E) Global human papillomavirus (HPV) phylogeny showing the clade containing the sequence recovered from the GIT of Pool 6 (highlighted in green). F) Mammalian orthoreovirus (MRV) phylogeny based on sequences recovered from the GIT of *D. rotundus*.

References

1. Chu DKW, Leung CYH, Gilbert M, et al. Avian Coronavirus in Wild Aquatic Birds. *J Virol.* 2011;85(23):12815–12820.
2. Moureau G, Temmam S, Gonzalez JP, et al. A real-time RT-PCR method for the universal detection and identification of flaviviruses. *Vector Borne Zoonotic Dis.* 2007;7(4):467-478.
3. Anthony SJ, Islam A, Johnson C., et al. Non-Random Patterns in Viral Diversity. *Nat Commun.* 2015;6(1):8147.
4. Heaton PR, Johnstone P, McElhinney LM, et al. Heminested PCR Assay for Detection of Six Genotypes of Rabies and Rabies-Related Viruses. *J Clin Microbiol.* 199;35(11):2762-2766.
5. Tong S, Chern SWW, Li Y, et al. Sensitive and Broadly Reactive Reverse Transcription-PCR Assays To Detect Novel Paramyxoviruses. *J Clin Microbiol.* 2008; 46(8):2652-2658.
6. Bourhy H, Cowley JA, Larrous F, et al. Phylogenetic Relationships among Rhabdoviruses Inferred Using the L Polymerase Gene. *J Gen Virol.* 2005;86(10):2849-2858.
7. Vilsker M, Moosa Y, Nooij S, et al. Genome Detective: an automated system for virus identification from high-throughput sequencing data. *Bioinformatics.* 2019;35(5):871-3.
8. Martin DP, Varsani A, Roumagnac P, et al. RDP5: A Computer Program for Analyzing Recombination in, and Removing Signals of Recombination from Nucleotide Sequence Datasets. *Virus Evol.* 2021;7(1):veaa087.

9. Stanton R, Wilkinson GW, Fox, JD. Analysis of human herpesvirus-6 IE1 sequence variation in clinical samples. *J Med Virol.* 2003;71(4):578-584.
10. Katoh K, Rozewicki J, Yamada KD. MAFFT online service: multiple sequence alignment, interactive sequence choice and visualization. *Brief Bioinform.* 2019;19;20(4):1160-1166.
11. Baele G, Ji X, Hassler GW, et al. BEAST X for Bayesian phylogenetic, phylogeographic and phylodynamic inference. *Nat Methods.* 2025;22:1653-1656.
12. Minh BQ, Schmidt HA, Chernomor O, et al. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Mol Biol Evol.* 2020;37(5):1530-1534.
13. Anisimova M, Gil M, Dufayard JF, et al. Survey of branch support methods demonstrates accuracy, power, and robustness of fast likelihood-based approximation schemes. *Syst Bio.* 2011;60(5):685-699.
14. Maddison WP, Maddison DR. Mesquite: a modular system for evolutionary analysis. Version 4.01. 2025. <https://www.mesquiteproject.org>
15. Olsen C, Qaadri K, Moir R, et al. Geneious R7: a bioinformatics platform for biologists. 2014. In International plant and animal genome conference XXII.
16. Danecek P, Bonfield JK, Liddle J, et al. Twelve years of SAMtools and BCFtools. *Gigascience.* 2021;10(2):giab008.
17. Sousa LLF, Guilardi MD, Martins JO, et al. Phylogenetic inferences reveal multiple intra- and interhost genetic diversity among bat rabies viruses circulating in northeastern Brazil. *One Health Outlook.* 2025;7(1):1.